

Amplifier

Why a signalling pathway has four steps instead of one, and why the number at the bottom is so much larger than the number at the top.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/amplifier/

1 Before you open anything

Answer from what you already know about epinephrine binding a G protein-coupled receptor. Commit to a number before the demonstration shows you one.

PREDICTION

Ten molecules of hormone bind ten receptors on one liver cell. The signal passes through four steps: receptor → G protein → cAMP → activated enzyme → product. Roughly how many product molecules does that cell end up making? Give an order of magnitude — tens, thousands, millions.

Write one sentence saying how you got that number — what you assumed each step does to the count.

2 Two pathways, the same input

Open the demonstration. The horizontal axis of the top plot is logarithmic: every gridline is a factor of ten, so equal spacing between bars means equal multiplication between tiers.

1. Press **Reset everything**. Check that **Hormone molecules bound** reads 10, the four tier factors read **×20**, **×100**, **×10**, **×1,000**, and **Terminator activity** reads **×1 normal**.
2. From the **Molecule count at each tier** table, copy the *Molecules* column into the first data column below.
3. Press **Bucket brigade**, which sets all four tiers to ×1. Copy the *Molecules* column again into the second data column.
4. Press **Realistic cascade** to restore the four factors.

Tier	Molecules — realistic cascade	Molecules — bucket brigade
Hormone bound		
Gα-GTP active		
cAMP		
Enzyme activated		
Product		

With the realistic cascade restored, record the four boxes under **The result**.

Product molecules from the cascade	Total gain	Bucket brigade would give	Cascade ÷ bucket brigade

a. The number of hormone molecules was 10 in both configurations, and nothing was added at any tier. Name the one property of the intermediates that differs between the two columns.

b. Add the four tier factors: $20 + 100 + 10 + 1,000$. Compare that total with the *Total gain* you recorded. Which arithmetic operation actually combines the tiers, and why does that operation follow from each step being catalytic?

The per-tier factors on this page are illustrative orders of magnitude — the size textbook cascades are roughly known to be, tens to thousands per tier. They are not measured constants for one named pathway in one named cell. Every total you record here is an order of magnitude, not a fact to quote.

3 What one tier is worth

Change one tier at a time and hold the rest still. This is the relationship the rest of the worksheet depends on.

1. Press **Realistic cascade**, and leave **Hormone molecules bound** at 10 and **Terminator activity** at **×1 normal**.
2. Drag slider **4. Product per activated enzyme** until the value beside it reads **×1**. Record the two readouts.

3. Repeat for $\times 10$, $\times 100$, $\times 1,000$ and $\times 10,000$. Every one of those values is on the slider — drag until the label matches exactly.

Tier 4 factor	Total gain	Product molecules from the cascade	Product $\div$ product in the row above
$\times 1$			—
$\times 10$			
$\times 100$			
$\times 1,000$			
$\times 10,000$			

- a. Write the rule in your own words: multiplying one tier's factor by ten does what to the number of product molecules?

- b. Write the total gain as an equation in the four tier factors g_1 to g_4 , then write the product count in terms of the gain and the number of hormone molecules bound, N_0 .

- c. Watch the *Product* bar as you change tier 4 by a factor of ten. How far does it move on the log axis, and why is that distance the same every time?

- d. Go back to your prediction in part 1. Was your order of magnitude right? If not, state specifically what you assumed a step does that the cascade does not do.

4 Working backwards, and a shortcut that fails

- A. **Designing a cascade to a target.** Work this out on paper first, then set it and check.

1. Set **Hormone molecules bound** to **50**, **Terminator activity** to **×1 normal**, tier 1 to **×10**, tier 2 to **×50** and tier 4 to **×500**.
 2. Leave tier 3 alone for the moment. On paper, find the tier 3 factor that would make **Product molecules from the cascade** read exactly **1,250,000,000**.
 3. Now set tier 3 to your answer and read the box.
- a. Show the working that gave your tier 3 factor.

B. The shortcut. Press **Reset everything** before starting this half.

b. Terminators are what end the signal: Ga hydrolysing its own GTP, phosphodiesterase destroying cAMP, phosphatases stripping phosphates back off. It is reasonable to expect that halving their activity, so every activated molecule survives twice as long, doubles the output. *Predict* the product count at **Terminator activity ×0.5** before you move anything.

c. Now set **Terminator activity** to **×0.5**, then to **×5**, then to **×10**. Record each product count and the factor by which it differs from the ×1 value.

Terminator activity	Product molecules from the cascade	Factor vs. the ×1 value
×1 normal		1
×0.5		
×5		
×10		

d. Each factor in your last column is a power of the terminator setting. What power, and why that power and not some other? Your answer must refer to how many tiers depend on how long an activated molecule survives.

e. The lower plot is a straight line on log–log axes. Using your answer to *d*, state its slope and explain in one sentence why a straight line of that slope is what a power relationship looks like on these axes.

5 Solve these without the demonstration

Close the page, or look away from it. Show your working, in scientific notation where it helps. Then reopen the demonstration, set it up, and check yourself — and if you were wrong, write down where it went off.

1. A four-tier cascade starts with 40 hormone molecules bound. The tier factors are $\times 50$, $\times 200$, $\times 5$ and $\times 2,000$, and the terminators are at normal speed. Find the total gain and the number of product molecules.

2. Take the standard cascade — 10 hormone molecules bound and factors $\times 20$, $\times 100$, $\times 10$, $\times 1,000$ — which delivers 2.0×10^8 product molecules at normal terminator speed. The terminators are now running five times faster. Find the new product count *without* recomputing tier by tier.

3. A cell needs 6.0×10^8 product molecules and has only 30 hormone molecules bound. Its first three tiers give $\times 20$, $\times 100$ and $\times 10$, and the terminators are at normal speed. What total gain is required, and what must the fourth tier factor be?

4. Someone proposes inserting a fifth protein into the pathway, one that is activated by the tier above it and activates exactly one molecule of the tier below. Explain what that insertion does to the number of product molecules, what it does to the time the response takes, and how the top plot would show it. No calculation is needed, but your answer must use the word *gain*.

6 On your own

Nothing here is graded. The demonstration does more than this worksheet uses.

- Drag **Hormone molecules bound** from 1 to 100 while watching both *Total gain* and *Product molecules from the cascade*. One of them moves and one does not. Explain which is which, then confirm it against the equation you wrote in part 3b.
- Using the lower plot, work out how fast the terminators would have to run to bring the cascade output down to the ten molecules a bucket brigade delivers. Check whether that setting exists on the slider, and say what that tells you about how hard it is to silence a cascade purely by speeding up termination.
- The hint under tier 3 says it takes four cAMP to release the catalytic subunits of one protein kinase A, yet the tier factor is a net $\times 10$. Set tier 3 to $\times 1$ and notice the slider stops there. Decide what a factor below one would have to mean physically, and whether a real step could have one.
- Turn off **Show the bucket-brigade comparison**, then set all four tiers to $\times 1$ by hand rather than with the button, and read the caption that appears under the written-out multiplication.